

AMAL JYOTHI
COLLEGE OF ENGINEERING
A U T O N O M O U S
KANJIRAPPALLY

ASSIGNMENT

23INMCA306 – Visual Programming

TITLE: SMART RETAIL BILLING AND
EMPLOYEE INCENTIVE MANAGEMENT
SYSTEM

Submitted To

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Smart Retail Billing and Employee Incentive Management System

1. VB.NET source code

Public Class Form1

Public Const GST_RATE As Double = 0.18 'Constant

Friend StoreName As String = "SMART RETAIL STORE"

Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load

lblTitle.Text = StoreName

End Sub

A

Private Sub btnGenerateBill_Click(sender As Object, e As EventArgs) Handles
btnGenerateBill.Click

Dim customerName As String

Dim itemCount As Integer

Dim total As Double = 0

Dim discount As Double

Dim gst As Double

Dim finalAmount As Double



'Input & Validation

customerName = UCase(Trim(txtCustomerName.Text))

itemCount = Val(txtItemCount.Text)

If customerName = "" Or itemCount <= 0 Then

 MsgBox("Please enter valid customer details", MsgBoxStyle.Exclamation)

 Exit Sub

End If

'Dynamic Array

Dim prices(itemCount - 1) As Double

'Loop for item prices

For i As Integer = 0 To itemCount - 1

 prices(i) = Val(InputBox("Enter price of item " & (i + 1)))

 total += prices(i)

Next

'Calculations

discount = CalculateDiscount(amount:=total)

gst = total * GST_RATE

finalAmount = total - discount + gst

'Output

MsgBox("Customer Name : " & customerName &

 vbCrLf & "Total Amount : ₹" & total &

 vbCrLf & "Discount : ₹" & discount &



```
vbCrLf & "GST : ₹" & gst &  
vbCrLf & "Final Bill : ₹" & finalAmount &  
vbCrLf & "Date : " & Date.Now,  
MsgBoxStyle.Information, "Customer Bill")
```

End Sub

```
Private Function CalculateDiscount(ByVal amount As Double,  
Optional loyalty As Boolean = False) As Double
```

```
Dim discount As Double = 0
```

```
If amount >= 5000 Then
```

```
discount = amount * 0.1
```

```
ElseIf amount >= 3000 Then
```

```
discount = amount * 0.05
```

```
End If
```

```
'Optional argument
```

```
If loyalty = True Then
```

```
discount += amount * 0.02
```

```
End If
```

```
Return discount
```

```
End Function
```



Private Sub btnCalculateIncentive_Click(sender As Object, e As EventArgs) Handles
btnCalculateIncentive.Click

Dim empId As Integer

Dim empName As String

Dim sales As Double

Dim incentive As Double

empId = Val(txtEmpId.Text)

empName = UCase(Trim(txtEmpName.Text))

sales = Val(txtSales.Text)

If empId <= 0 Or empName = "" Or sales <= 0 Then

MsgBox("Please enter valid employee details", MsgBoxStyle.Exclamation)

Exit Sub

End If

incentive = CalculateIncentive(sales)

MsgBox("Employee ID : " & empId &

vbCrLf & "Employee Name : " & empName &

vbCrLf & "Sales Amount : ₹" & sales &

vbCrLf & "Incentive : ₹" & incentive,

MsgBoxStyle.Information, "Employee Incentive")

End Sub



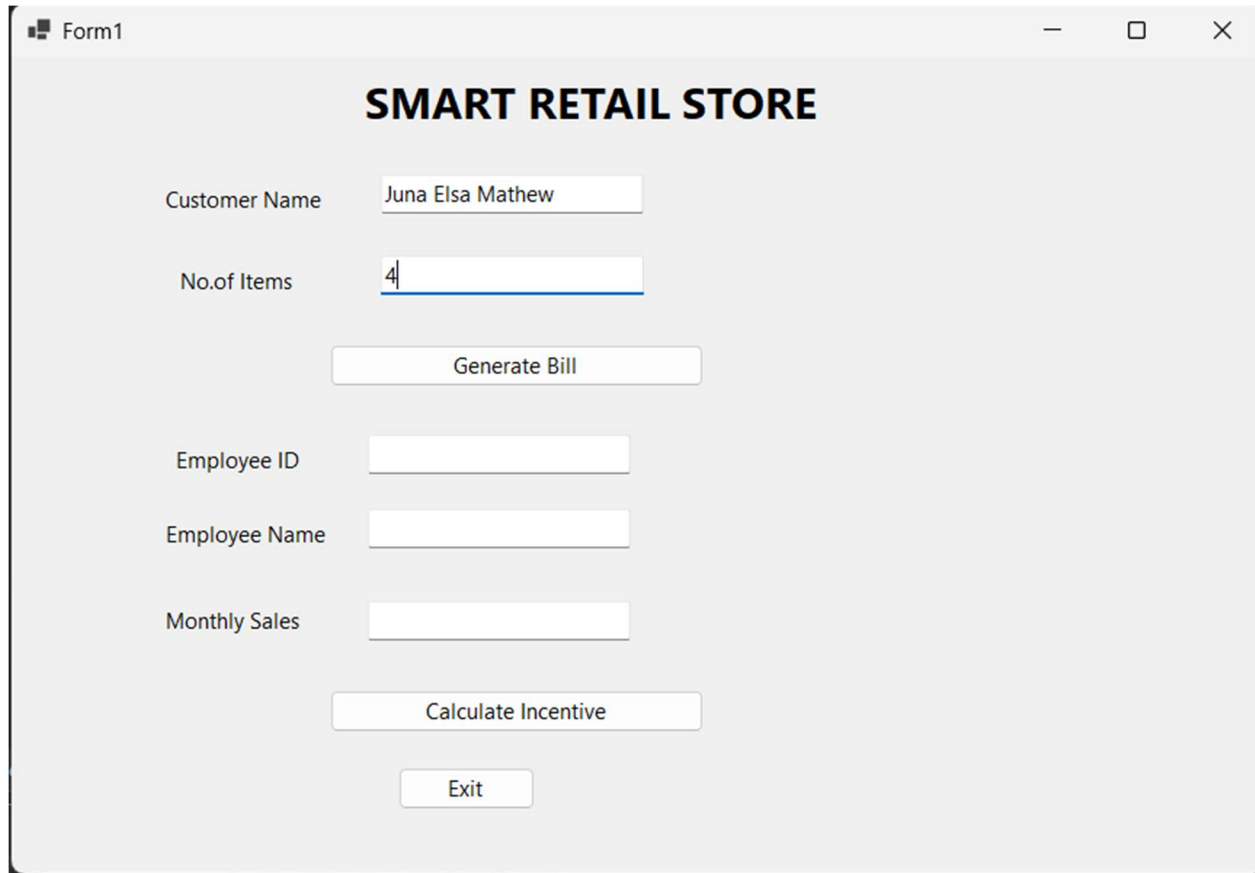
```
Private Function CalculateIncentive(ByRef sales As Double) As Double
    If sales >= 100000 Then
        Return sales * 0.15
    ElseIf sales >= 50000 Then
        Return sales * 0.1
    Else
        Return sales * 0.05
    End If
End Function

Private Sub btnExit_Click(sender As Object, e As EventArgs) Handles btnExit.Click
    End
End Sub

End Class
```



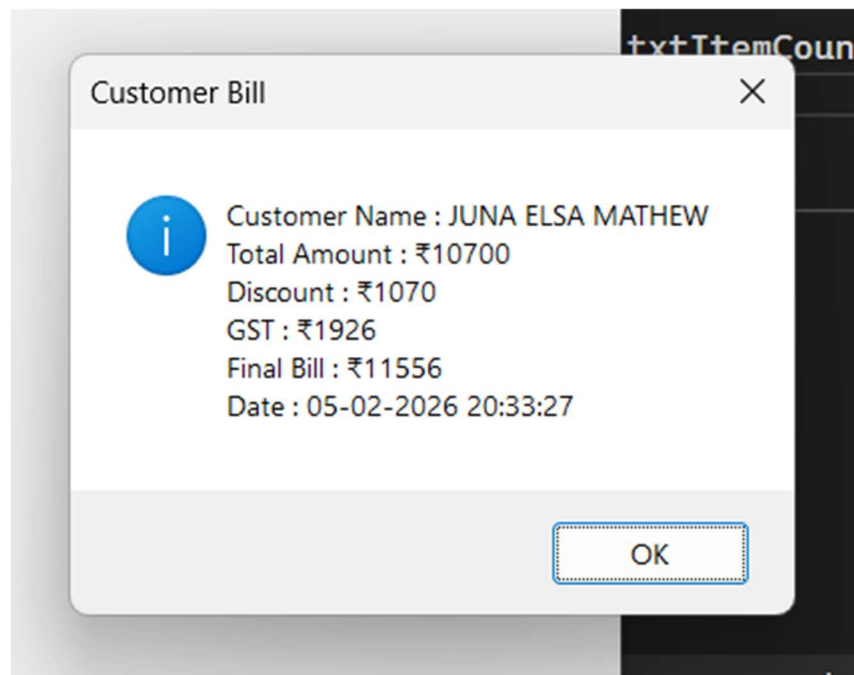
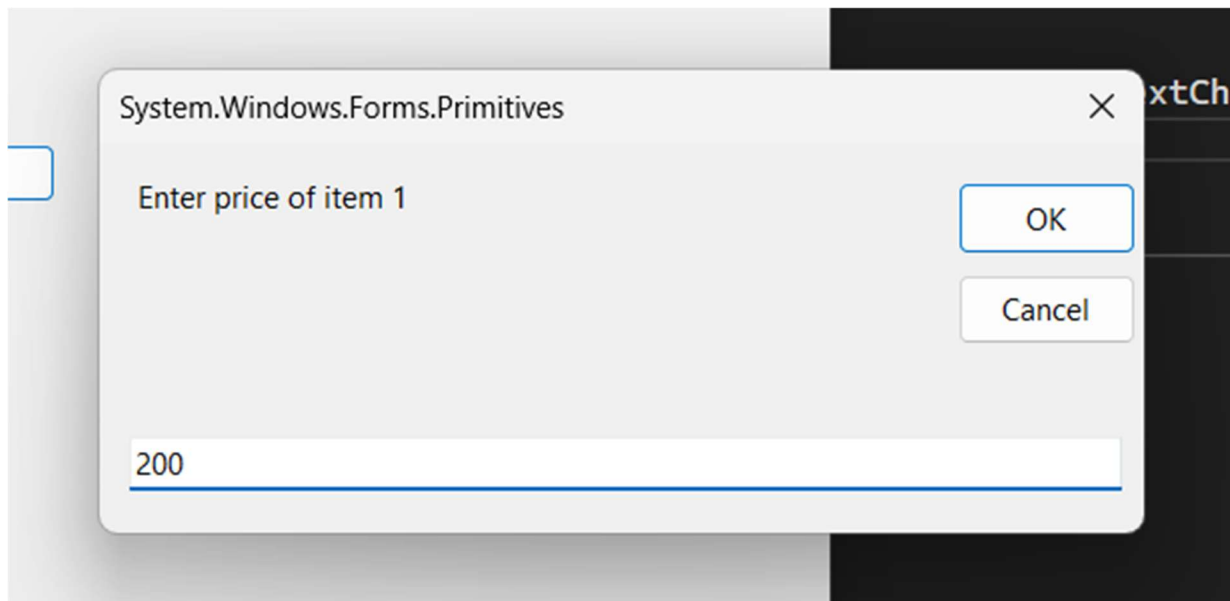
2. Output screenshots



The screenshot displays a Windows application window titled "Form1". The main heading is "SMART RETAIL STORE". Below the heading, there are several input fields and buttons:

- Customer Name:** A text box containing "Juna Elsa Mathew".
- No. of Items:** A text box containing "4".
- Generate Bill:** A button located below the "No. of Items" field.
- Employee ID:** A text box.
- Employee Name:** A text box.
- Monthly Sales:** A text box.
- Calculate Incentive:** A button located below the "Monthly Sales" field.
- Exit:** A button located at the bottom center of the form.





Form1

SMART RETAIL STORE

Customer Name

No.of Items

Employee ID

Employee Name

Monthly Sales

Employee Incentive

 Employee ID : 101
Employee Name : ELSA MATHEW
Sales Amount : ₹40000
Incentive : ₹2000



REPORT

Smart Retail Billing and Employee Incentive Management System

1. Problem Definition

A medium-sized retail store requires an automated system to manage its customer billing process and employee incentive calculation. Manual billing is time-consuming and error-prone, especially when handling multiple items, discounts, GST calculations, and incentive slabs. To overcome these issues, a VB.NET Windows Forms–based application is developed to efficiently generate customer bills and compute employee incentives based on monthly sales.

The system provides a user-friendly interface to accept customer and employee details, perform accurate calculations, and display formatted outputs. It also demonstrates key VB.NET programming concepts as prescribed in the syllabus.

2. Module Description

The application consists of the following main modules:

2.1 Customer Billing Module

This module is responsible for generating customer bills.

- Accepts customer name and number of items purchased
- Stores item prices in a dynamic array
- Calculates total amount
- Applies discount based on purchase amount
- Calculates GST using a constant rate
- Displays the final bill using a message box along with date and time

2.2 Employee Incentive Module

This module calculates incentives for employees based on their monthly sales.

- Accepts employee ID, name, and sales amount
- Calculates incentive using a function procedure
- Uses pass-by-reference for sales amount
- Displays incentive details in a formatted message box



2.3 Menu / Control Module

- Provides buttons to select Customer Billing or Employee Incentive
- Allows repeated operations until the user exits
- Ensures smooth navigation within the application

3. Concepts Used

The project demonstrates the following VB.NET concepts:

- **Variables and Data Types:** String, Integer, Double
- **Constants:** GST rate declared using Const
- **Arrays:** Dynamic array to store item prices
- **Operators:** Arithmetic and relational operators
- **Control Structures:** If-Else, For loop
- **Procedures:** Sub procedures for input and display
- **Functions:** Used for discount and incentive calculations
- **Parameter Passing:**
 - Pass By Value (Discount calculation)
 - Pass By Reference (Incentive calculation)
 - Optional arguments (Loyalty discount)
 - Named arguments
- **Scope and Access Control:**
 - Public constants
 - Friend variables
 - Private functions
- **Predefined Functions:**
 - String functions: UCase, Trim, Len
 - Date and time function: Date.Now
- **Windows Forms Controls:** Labels, TextBoxes, Buttons, MessageBox

4. Sample Output

4.1 Customer Billing Output

Customer Name : ARUN
 Total Amount : ₹5200
 Discount : ₹520
 GST : ₹936
 Final Bill : ₹5616
 Date : 02-02-2026 10:30 AM



4.2 Employee Incentive Output

Employee ID : 101

Employee Name : SNEHA

Sales Amount : ₹75000

Incentive : ₹7500

5. Conclusion

The Smart Retail Billing and Employee Incentive Management System successfully automates billing and incentive calculations using VB.NET Windows Forms. The project fulfills all functional and non-functional requirements and demonstrates essential programming concepts such as modularity, arrays, parameter passing, and access control. This application can be further enhanced by adding database connectivity and report generation features in the future.

